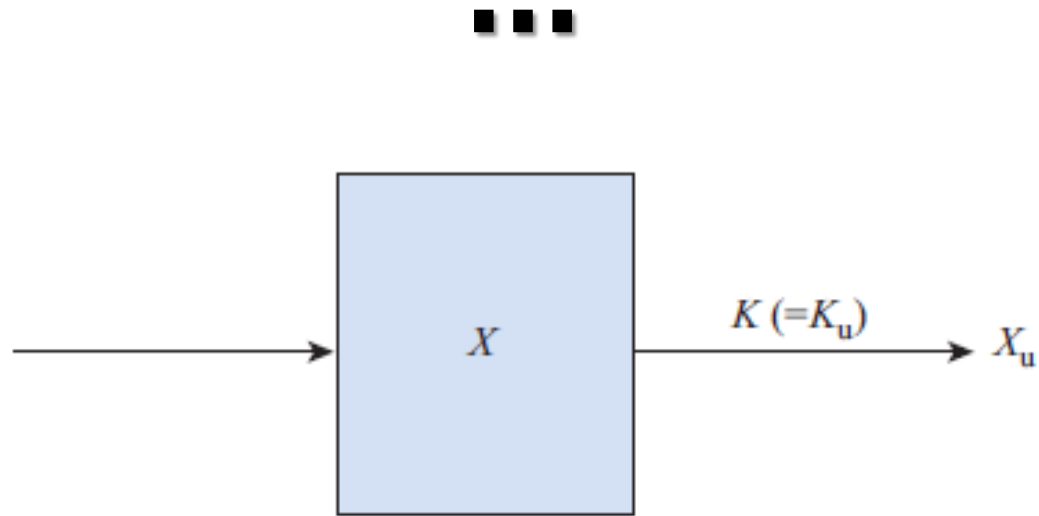


2.5 PK of IV bolus administration



- Assumption

- One-compartment model, first-order process and passive diffusion are operative,
- No metabolism takes place (elimination is 100% via renal excretion),
- The drug is being monitored in blood (plasma/serum) and urine.



- Where,
 - X , mass (amount) of drug in the blood/body at time, t ;
 - X_u , mass (amount) of unchanged drug in the urine at time, t ;
 - K , first-order elimination rate constant.



- Useful pharmacokinetic parameters include:
 - Apparent volume of distribution
 - Elimination half life
 - Elimination rate constant
 - Systemic clearance.



- The above parameters are computed using data either from:
 - Drug concentrations in the plasma, or
 - Urinary excretion data



Drug concentrations in the Plasma

- Following rapid IV injection of a drug that distributes in the body according to a one-compartment model and is eliminated by apparent first-order kinetics,
 - The rate of loss of drug from the body is given by:

$$\frac{dX}{dt} = -KX$$

■ ■ ■

- The integration of the above equation gives;

$$X = X_0 e^{-Kt}$$

- And dividing both sides by a volume term;

$$\frac{X}{V} = \frac{X_0}{V} e^{-Kt}$$

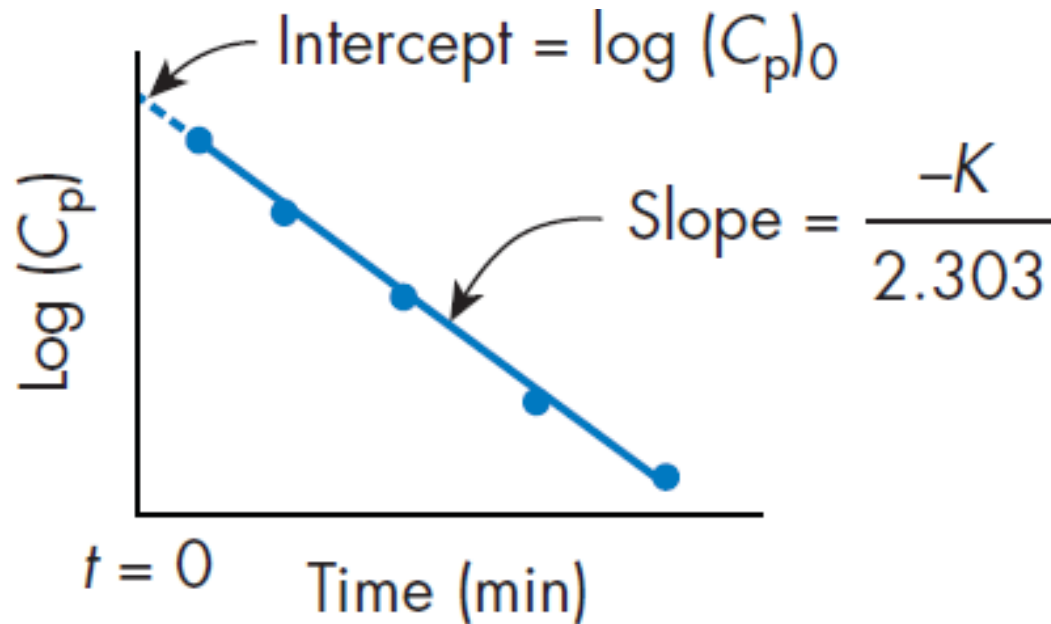
- Yields;

$$C = C_0 e^{-Kt}$$

■ ■ ■

1. Apparent volume of distribution (V)

- A plot of ($\log C$ vs. t) will be linear under the conditions stated.





- C_o may be obtained by extrapolation of $\log C$ vs. t plot to zero.
- This intercept, C_o , may be used in the calculation of the apparent volume of distribution.
- Since X_o equals the amount of drug injected IV (i.e., IV dose), then:

$$V = \frac{X_o}{C_o} = \frac{IV \text{ dose}}{C_o}$$



2. Elimination half life ($t_{1/2}$)

- $t_{1/2}$ is readily determined from a semi-logarithmic plot of C_p vs t .
- The Eqn. $t_{1/2} = \frac{0.693}{K}$ is easily derived by setting $C = C_0/2$ and $t = t_{1/2}$ in ' $\log C = \log C_0 - \frac{Kt}{2.303}$ '



3. Elimination rate constant (K)

- The slope of the line resulting from a plot of $\log C$ vs. t is equal to $-\frac{Kt}{2.303}$ and K may be estimated directly from this slope.
- K can also be obtained rearranging the equation:

$$\log C = \log C_0 - \frac{Kt}{2.303}$$

- It is easier, however, to estimate K from the relationship:

$$K = \frac{0.693}{t_{1/2}}$$



Drug concentrations in the Urine

- It is sometimes possible to determine the elimination kinetics of a drug from urinary excretion data.
- This requires that at least some of the drug be excreted **unchanged**.

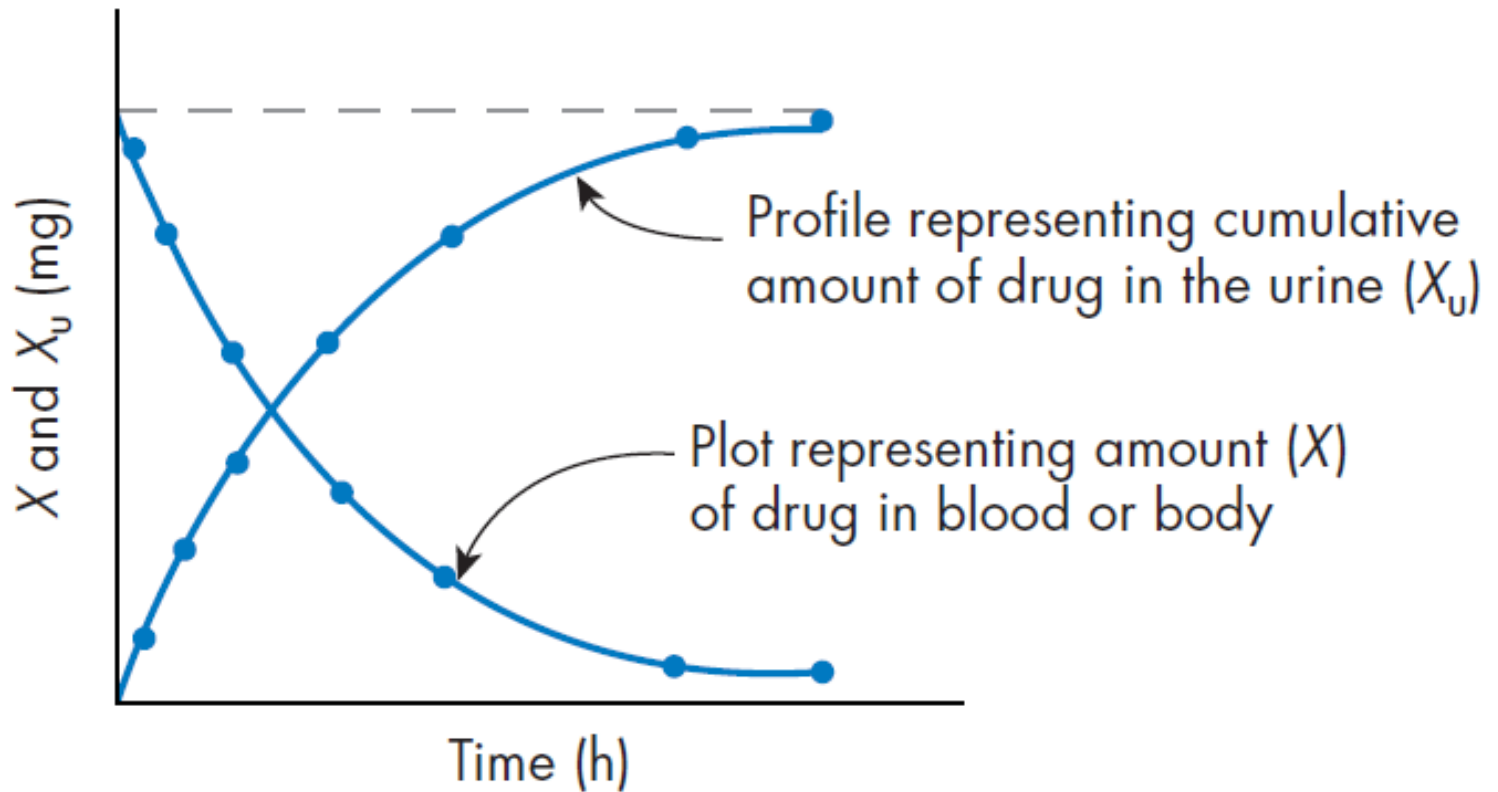


Fig. Plot illustrating the amount of drug remaining in the blood and the amount of drug eliminated in the urine with time following the administration of a drug as an intravenous bolus.



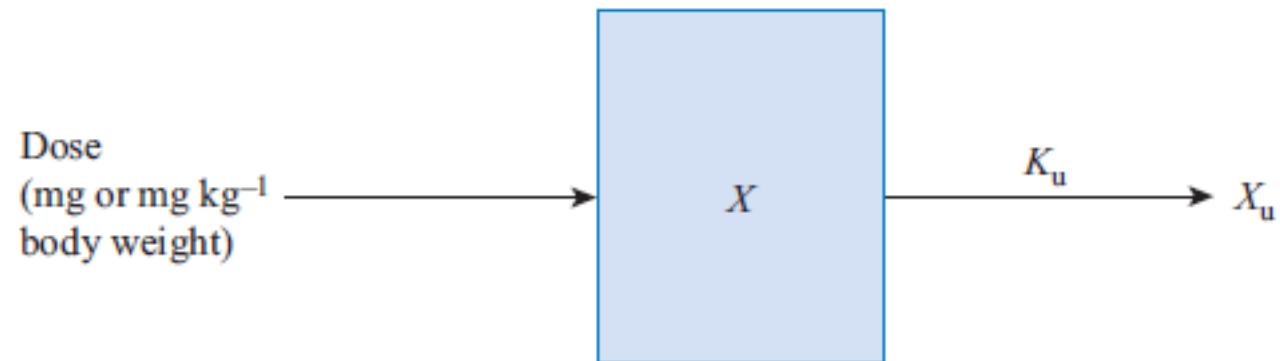
- Advantages

- Urine collection is a non-invasive technique.
- A more convenient method of sample collection, and sample size is generally not a problem.
- Urinary data allows direct measurement of bioavailability.

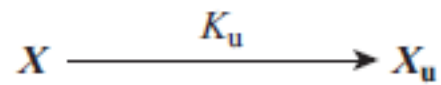


- Assumptions:
 - One-compartment open model with the entire dose eliminated as unchanged drug
 - First-order process and passive diffusion
 - $(X_u)_\infty = X_o$ (at $t = \infty$)
 - $K = K_u$ (drug is exclusively eliminated by excretion by kidney)

■ ■ ■



SETUP





- There are **two methods** that permit us to compute some pharmacokinetic parameters from urinary excretion data.
 1. The rate of excretion method.
 2. The “amount remaining to be excreted” method (ARE); also known as the sigma-minus method



Rate of excretion method

- The rate of appearance of intact drug in the urine is proportional to the amount of drug in the body,

$$\frac{dX_u}{dt} = K_u X$$

Where, $\frac{dX_u}{dt}$ = rate of excretion, X is the amount of drug in the body at time t , X_u is the cumulative amount of drug excreted into urine at time t , and K_u is the excretion/elimination rate constant

■ ■ ■

- Substitution for X according to Eqn. $X = X_0 e^{-Kt}$ yields

$$\frac{dX_u}{dt} = K_u X_0 e^{-Kt}$$

- Therefore, taking the log of the above Eq.,

$$\log \frac{dX_u}{dt} = \log K_u X_0 - \frac{Kt}{2.303}$$

- Note, $K=K_u$

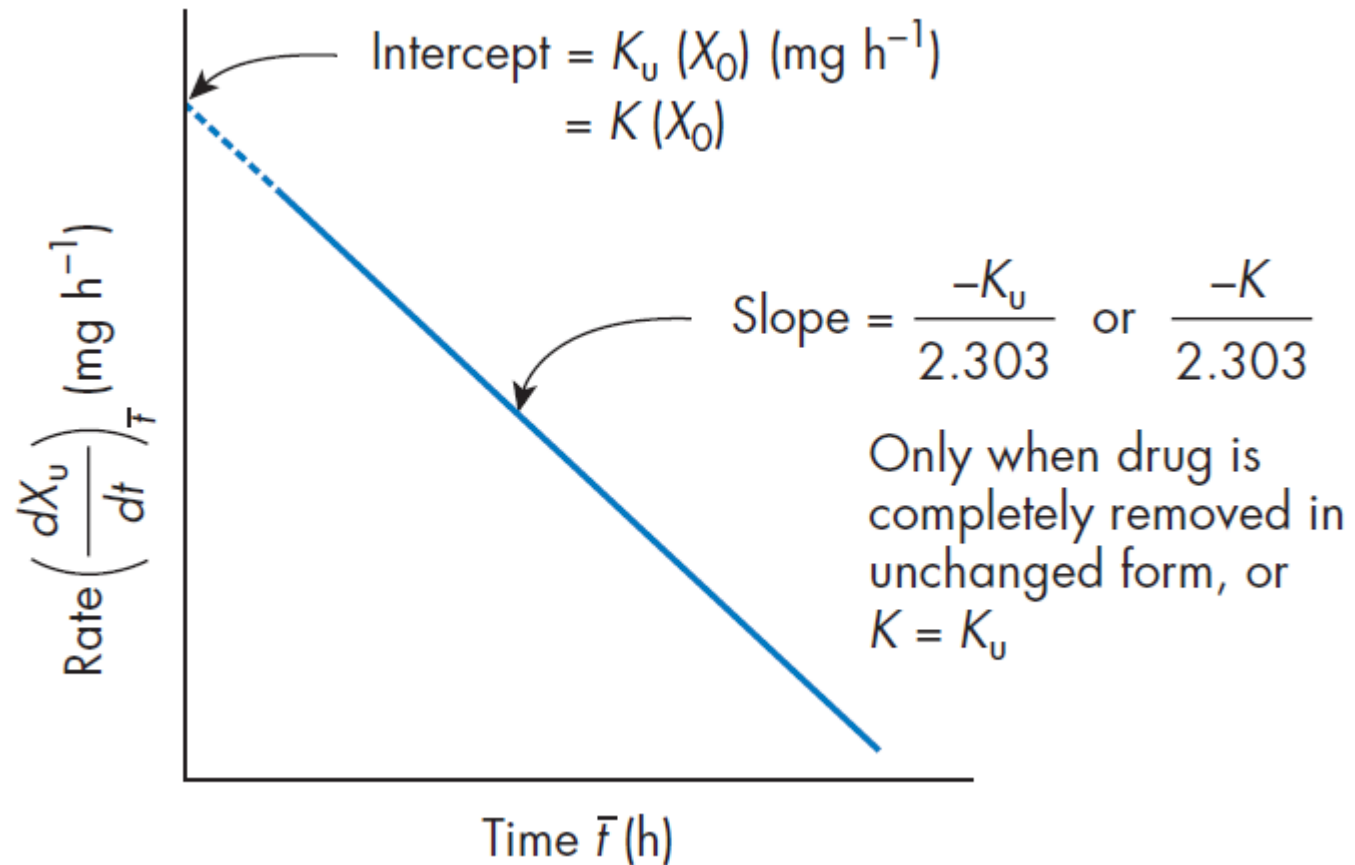


Fig. Semi log plot of excretion rate vs. t after IV administration of a drug.



- Eqn. states that a semi log plot of excretion rate of un-metabolized drug versus time is linear, with a slope of $-\frac{K}{2.303}$.
- This slope is the same as that obtained from a semi log plot of C_p vs. t .
- Thus, the elimination rate constant of a drug can be obtained from either plasma concentration or urinary excretion data.



- Urinary excretion rates are estimated by:
 - Collecting all urine for a fixed period of time,
 - Determining the concentration of drug in the urine,
 - Multiplying the concentration by the volume of urine collected to determine the amount excreted, and
 - Dividing the amount excreted by the collection time.



- Table: Calculation of excretion rate vs. time data for estimating half-life.

t (h)	X_u (mg)	Δt	ΔX_u	$\Delta X_u / \Delta t$ (mg/h)	t_m
0	0.0	1	4.0	4.0	0.5
1	4.0	1	3.8	3.8	1.5
2	7.8	1	3.5	3.5	2.5
3	11.3	3	9.1	3.0	4.5
6	20.4	6	13.5	2.2	9.0
12	33.9	12	14.7	1.2	18.0
24	48.6	12	6.4	0.53	30.0
36	55.0	12	2.8	0.23	42.0
48	57.8				



- These experimentally determined excretion rates are obviously not instantaneous rates (i.e., $\frac{dX_u}{dt}$), but are average rates over a finite time period (i.e., $\frac{\Delta X_u}{\Delta t}$).
- However, we often find that $\frac{\Delta X_u}{\Delta t}$ closely approximates $\frac{dX_u}{dt}$ at the midpoint of the urine collection period.



- Limitation of the rate of excretion method
 - Fluctuations in the rate of drug elimination are reflected to a high degree in excretion rate plots.
 - At times, the data are so scattered that an estimate of the half-life is difficult.
- To overcome this problem, an alternative approach, termed the **sigma-minus method**, is available.

■ ■ ■

The amount remaining to be excreted (sigma minus) method

$$\frac{dX_u}{dt} = K_u X$$

Where, X is the amount of drug in the body at time t , X_u is the cumulative amount of drug excreted into urine at time t , and K_u is the excretion/elimination rate constant



- Integrating the above equation ($\frac{dX_u}{dt} = K_u X$)

$$X_O - (X_u)_t = X_O e^{-K_u t}$$

- When $t = \infty$, $(X_u)_\infty = X_O$ (dose)

$$(X_u)_\infty - (X_u)_t = X_O e^{-K_u t}$$

- $(X_u)_\infty - (X_u)$ is the amount of drug remaining to be excreted (ARE)
- A plot of ARE vs. t is a straight line

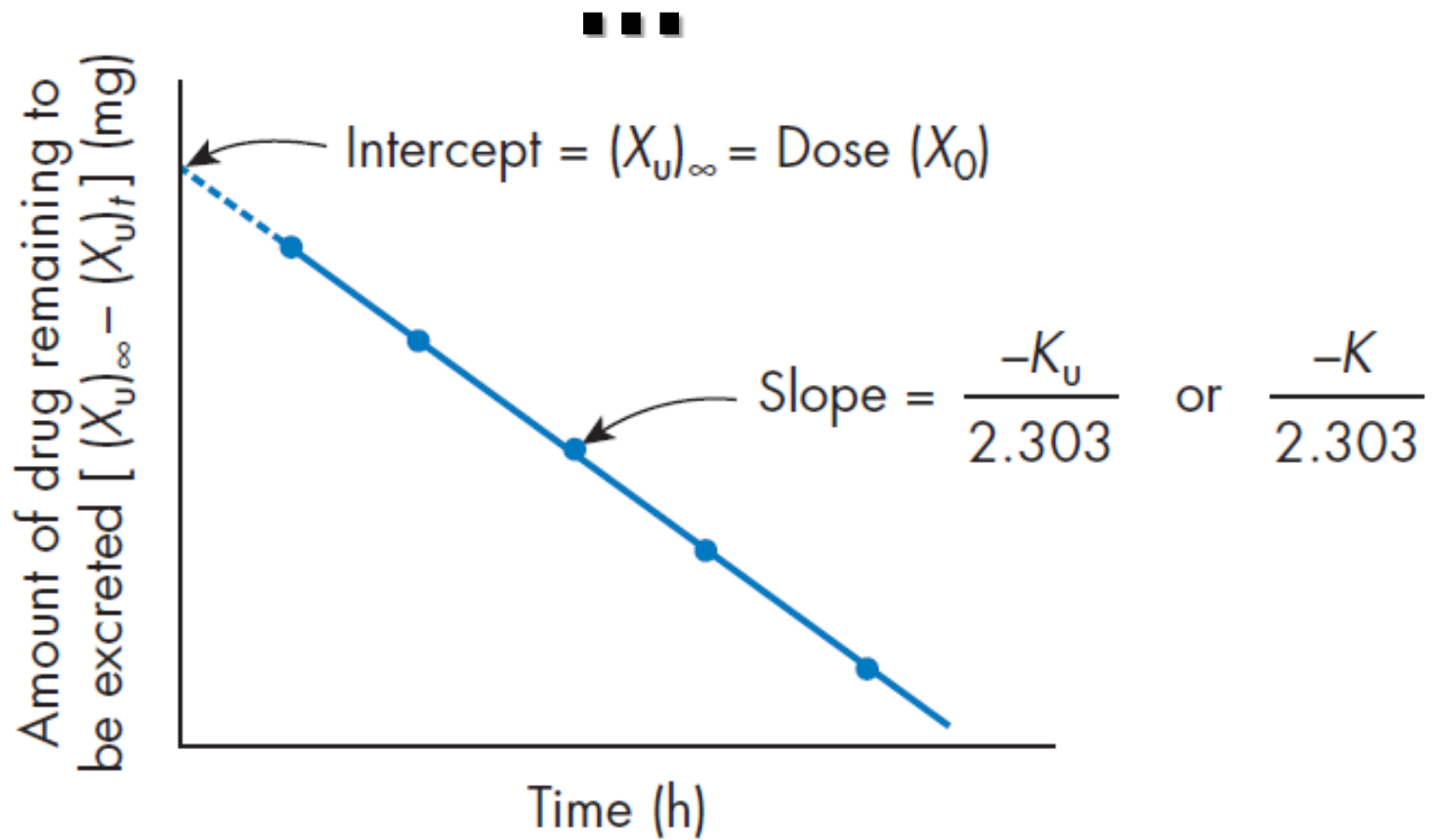


Fig. Semi log plot of ARE vs. t after IV administration of a drug (On a semilog paper).



- The slope of the graph permits the determination of the excretion rate constant which is equal to the elimination rate constant.
 - We can also obtain the elimination half life
 - The intercept is $(X_u)_\infty$ or X_0 (dose).
 - This is true only when there is an absence of metabolite(s)



- Limitations of the ARE method:
 - Urine samples must be collected until such time that, practically, no additional drug appears in the urine (i.e. $t = 7 t_{1/2}$)
 - No urine samples can be lost (the exact volume of urine at each time interval must be known)
 - This is a time-consuming (especially for a drug with a long $t_{1/2}$)
 - There is a cumulative build up of error.



- In general, elimination rate constant may be estimated from plots of
 - log drug concentration in plasma versus time,
 - log excretion rate versus time (the rate method), and
 - log A.R.E. versus time (the sigma-minus method).



Estimating Renal Clearance

- The kinetics of renal excretion of a drug may be characterized not only by a renal excretion rate constant K_u , but also by a renal clearance CL_R .
- By definition, CL_R is simply the ratio of urinary excretion rate to drug concentration in the blood or plasma:

$$CL_R = \frac{dX_u/dt}{C} = \frac{K_u X}{C}$$



- Recognizing that $\frac{X}{C}$ is simply the apparent volume of distribution we can write:

$$CL_R = K_u \times V$$

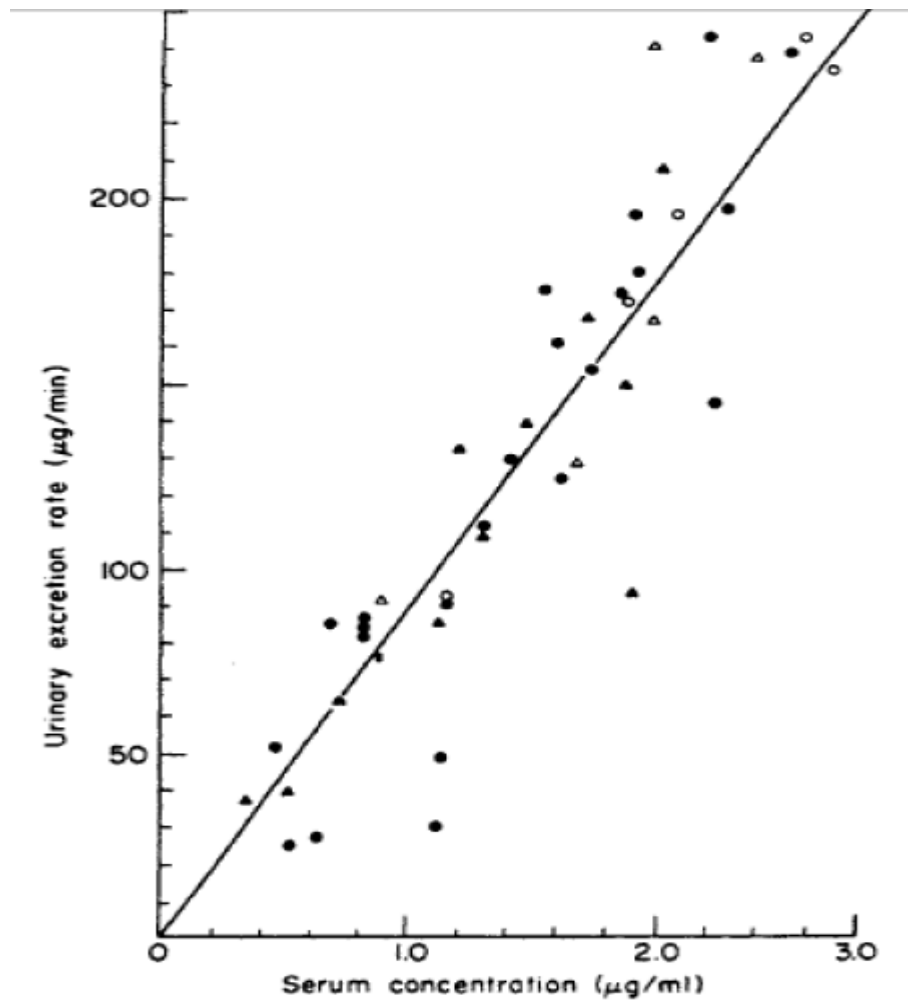
- An estimation of CL_R by means of ' $\frac{dX_u/dt}{C}$ ' may be misleading because like all rate processes in the body, renal excretion is subject to biologic variability.



- A more satisfactory approach is to plot $\frac{dX_u}{dt}$ vs. C at the midpoints of the urine collection periods.
- Since,

$$\frac{dX_u}{dt} = CL_R \times C$$

- The slope of an excretion rate-plasma concentration plot is equal to renal clearance.





Systemic Clearance

- It has been shown that the product of the urinary excretion rate constant K_u and V is equal to renal clearance.
- The product of the elimination rate constant K and V also yields a clearance term, **systemic clearance** (CL_S).
- It can be shown that the systemic clearance is given by the ratio of the IV dose to the AUC.

■ ■ ■

- Since,

$$CL_R = K_u \cdot V = \frac{X_u^\infty}{AUC}$$

- Then,

$$V = \frac{X_u^\infty}{K_u \times AUC}$$

- And, at $t = \infty$, $X_u^\infty = X_0$, and for a drug excreted only renally,
 $K_u = K$

■ ■ ■

- Therefore,

$$V = \frac{X_o}{K \times AUC} = \frac{CL_s}{K}$$

- And,

$$CL_s = V \cdot K$$